

Product Requirements Document (PRD)

Hotel Booking Analysis - Understanding Booking Cancellations

1. Document Overview

Field	Details
Document Title	Hotel Booking Analysis - PRD
Project Title	Hotel Booking Analysis
Document Type	Product Requirements Document (PRD)
Prepared For	Data Analyst / Analytics Team
Prepared By	Business Analyst / Project Manager
Document Purpose	To define business requirements, analytical scope, KPIs, and deliverables for the Hotel Booking Analysis project
Tools in Scope	Microsoft Excel, SQL, Power BI

This document defines **what** needs to be analyzed and **why**, so that a Data Analyst can independently plan and execute the project. It does not prescribe technical solutions, queries, formulas, or analysis outcomes.

2. Project Background

Hotels regularly receive bookings through multiple channels, and a portion of these bookings are cancelled before the guest's arrival. Cancellations affect room inventory planning, revenue predictability, and operational efficiency. Understanding booking behaviour and cancellation patterns is essential for the hotel to make informed business decisions.

3. Business Problem

A significant number of hotel bookings are being cancelled. This creates uncertainty in occupancy planning, affects revenue expectations, and impacts overall hotel performance. The business currently lacks a structured understanding of booking behaviour and the circumstances under which cancellations occur.

4. Problem Statement

Hotel Booking Cancellation — The hotel needs to understand booking and cancellation patterns to reduce avoidable cancellations and improve business planning.

5. Project Description

This project involves studying historical hotel booking data to understand:

- How customers book
- When and why cancellations occur
- How booking behaviour varies across customer types, channels, and hotel types
- How cancellations affect hotel performance

The output will support business decision-making, not automated prediction.

6. Business Objective

To analyze historical hotel booking data in order to identify patterns and business factors associated with booking cancellations, so the hotel can:

- Understand customer booking behaviour
 - Improve booking and inventory management
 - Reduce avoidable cancellations
 - Support better business decisions
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7. Business Goals

Goal	Description
Improve visibility	Provide clear visibility into booking and cancellation trends
Identify risk areas	Highlight booking segments with higher cancellation tendency
Support planning	Enable better forecasting of occupancy and revenue
Reduce business impact	Support initiatives to reduce cancellation-related losses
Enable data-driven decisions	Replace assumptions with data-backed business understanding

8. Project Scope

The project scope covers descriptive and diagnostic analysis of historical hotel booking data using Excel, SQL, and Power BI. It focuses on identifying patterns and associations related to cancellations — not predicting future cancellations.

9. In-Scope Requirements

- Analysis of booking volume and trends
- Analysis of cancellation rate and patterns
- Analysis of customer booking behaviour
- Analysis of booking channels

- Analysis of hotel type performance
 - Analysis of lead time, stay duration, and deposit-related factors
 - Analysis of seasonal booking patterns
 - Business impact assessment of cancellations
 - Dashboard reporting via Power BI
 - Data validation and preparation using Excel and SQL
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10. Out-of-Scope Requirements

- Predictive modelling or forecasting using Machine Learning
 - Use of Python, R, or advanced statistical techniques
 - Root-cause determination through experimentation (e.g., A/B testing)
 - Automated cancellation prediction systems
 - Real-time data pipelines
 - Customer-level personalization or marketing automation
 - Financial audit or accounting-level revenue reconciliation
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11. Stakeholders

Stakeholder	Role/Interest
Hotel Management / Leadership	Business decision-making, strategy
Revenue Management Team	Occupancy and pricing planning
Operations Team	Front-desk and inventory management
Sales & Booking Team	Understanding channel performance
Business Analyst	Requirement definition and validation
Data Analyst	Analysis execution and reporting

12. Target Users

- Hotel Management (strategic decisions)
 - Revenue and Operations Managers (planning decisions)
 - Sales/Booking Teams (channel and customer insights)
 - Business Analysts (ongoing monitoring)
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13. Business Questions

The Data Analyst is expected to investigate the following business questions:

1. What is the overall booking volume and how has it changed over time?

2. What percentage of bookings are cancelled?
3. How does the cancellation rate vary across hotel types?
4. Is there a relationship between booking lead time and cancellation likelihood?
5. Do certain booking channels show higher cancellation tendencies?
6. Does customer type (e.g., new vs. repeat) relate to cancellation behaviour?
7. How does length of stay relate to cancellations?
8. Do deposit or payment conditions relate to cancellation behaviour?
9. Does prior booking/cancellation history relate to future cancellations?
10. Do special requests relate to booking commitment or cancellation likelihood?
11. Are there seasonal or monthly patterns in bookings and cancellations?
12. What is the estimated business impact (e.g., lost room-nights) of cancellations?
13. Which segments (hotel type, channel, customer type) contribute most to cancellations?

Note: These are investigation areas for the Data Analyst. No relationships or causes should be assumed in advance.

14. Key Analytical Requirements

The analysis must be **descriptive and diagnostic** in nature (explaining what happened and identifying associated factors), not predictive.

Required analysis areas:

- Overall booking volume
- Cancellation rate
- Booking trends over time
- Customer booking behaviour
- Lead time vs. cancellation behaviour
- Booking channel vs. cancellation behaviour
- Customer type vs. cancellation behaviour
- Hotel type vs. cancellation behaviour
- Stay duration vs. cancellation behaviour
- Deposit/payment-related factors
- Previous booking history
- Special requests and booking characteristics
- Revenue/business impact of cancellations
- Seasonal booking and cancellation patterns

Each area should be treated as a **question to investigate**, not a confirmed relationship.

15. Key Performance Indicators (KPIs)

KPI Name	Purpose
Total Bookings	Measure overall booking volume
Cancellation Rate	Measure proportion of bookings cancelled
Confirmed Booking Rate	Measure proportion of bookings honoured
Average Lead Time	Understand booking-to-arrival time gap
Average Length of Stay	Understand typical stay duration
Repeat Guest Rate	Understand customer loyalty patterns
Revenue Impact of Cancellations	Understand business impact of cancellations
Channel-wise Booking Share	Understand contribution of each booking channel
Seasonal Booking Distribution	Understand demand patterns across periods

16. KPI Definitions and Business Purpose

16.1 Total Bookings

- **Definition:** Count of all bookings made in a given period
- **Purpose:** Understand overall demand
- **Dimensions:** Time period, hotel type, channel
- **Business Use:** Demand tracking

16.2 Cancellation Rate

- **Definition:** Proportion of bookings that were cancelled
- **Purpose:** Understand the scale of the cancellation problem
- **Dimensions:** Hotel type, channel, customer type, time period
- **Business Use:** Identify high-risk segments

16.3 Confirmed Booking Rate

- **Definition:** Proportion of bookings that resulted in a stay
- **Purpose:** Measure booking reliability
- **Dimensions:** Hotel type, channel, time period
- **Business Use:** Track booking quality

16.4 Average Lead Time

- **Definition:** Average time between booking date and arrival date
- **Purpose:** Understand planning behaviour of customers

- **Dimensions:** Hotel type, customer type, cancellation status
- **Business Use:** Booking pattern analysis

16.5 Average Length of Stay

- **Definition:** Average number of nights booked
- **Purpose:** Understand stay patterns
- **Dimensions:** Hotel type, season, customer type
- **Business Use:** Occupancy planning

16.6 Repeat Guest Rate

- **Definition:** Proportion of bookings made by repeat guests
- **Purpose:** Understand customer loyalty and its relation to cancellations
- **Dimensions:** Hotel type, channel
- **Business Use:** Loyalty and retention insight

16.7 Revenue Impact of Cancellations

- **Definition:** Estimated business impact resulting from cancelled bookings
- **Purpose:** Quantify the cost of cancellations
- **Dimensions:** Hotel type, time period, channel
- **Business Use:** Business case for reducing cancellations

16.8 Channel-wise Booking Share

- **Definition:** Distribution of bookings across booking channels
- **Purpose:** Understand channel performance
- **Dimensions:** Channel, hotel type
- **Business Use:** Channel strategy input

16.9 Seasonal Booking Distribution

- **Definition:** Distribution of bookings across months/seasons
- **Purpose:** Understand demand seasonality
- **Dimensions:** Month, hotel type
- **Business Use:** Seasonal planning

17. Required Business Dimensions and Metrics

Dimensions:

- Hotel Type
- Booking Channel
- Customer Type
- Time Period (Year, Month, Week, Day)
- Deposit/Payment Type
- Special Request Indicator

- Repeat Guest Indicator

Metrics:

- Booking Count
 - Cancellation Count
 - Cancellation Rate
 - Lead Time
 - Length of Stay
 - Revenue/Business Impact Value
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18. Functional Requirements

- The analysis must allow filtering by hotel type, channel, customer type, and time period.
 - The analysis must support comparison across categories (e.g., channel vs. cancellation rate).
 - The analysis must support trend analysis over time.
 - The output must clearly separate cancelled vs. non-cancelled bookings for comparison.
 - The analysis must support summary-level and detail-level views.
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19. Non-Functional Requirements

- The analysis and dashboard must be easy to understand for non-technical business users.
 - Reports must be visually clear and business-friendly.
 - Performance of dashboards should be reasonable for the given data volume.
 - Terminology used must be simple and consistent throughout.
 - Analysis must be reproducible and well-documented.
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20. Data Requirements

The following categories of data may be required (no dataset should be created or assumed):

- **Booking Information:** Booking date, arrival date, booking status
- **Customer Information:** Customer type, repeat guest indicator
- **Hotel Information:** Hotel type/category
- **Stay Information:** Length of stay, number of guests
- **Cancellation Information:** Cancellation status, cancellation date (if available)
- **Booking Channel Information:** Source/channel of booking
- **Payment/Deposit Information:** Deposit type or payment condition
- **Revenue-Related Information:** Rate/price-related fields, where available

21. Data Quality Requirements

- Data must be checked for missing or blank values in key fields.
 - Data must be checked for duplicate booking records.
 - Date fields must be validated for logical consistency (e.g., arrival date after booking date).
 - Categorical fields must be checked for inconsistent labelling (e.g., inconsistent naming of channels).
 - Any data quality issues must be documented before analysis begins.
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22. Expected Analysis Areas

- Booking volume and trend analysis
 - Cancellation rate analysis
 - Customer behaviour analysis
 - Channel performance analysis
 - Hotel type performance analysis
 - Lead time analysis
 - Length of stay analysis
 - Deposit/payment behaviour analysis
 - Repeat guest behaviour analysis
 - Special request analysis
 - Seasonal trend analysis
 - Business impact analysis of cancellations
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23. Power BI Dashboard Requirements

23.1 Dashboard Pages/Sections

1. **Overview Page** – overall booking volume, cancellation rate, key KPI cards
2. **Cancellation Analysis Page** – cancellation patterns by channel, hotel type, customer type
3. **Customer Behaviour Page** – lead time, repeat guests, special requests
4. **Trend & Seasonality Page** – booking and cancellation trends over time
5. **Business Impact Page** – revenue/business impact of cancellations

23.2 KPI Cards

- Total Bookings
- Cancellation Rate
- Confirmed Booking Rate
- Average Lead Time
- Revenue Impact of Cancellations

23.3 Required Visual Categories

- Trend charts (line/area) for bookings and cancellations over time
- Bar/column charts for comparison across categories (channel, hotel type, customer type)
- Distribution charts for lead time and length of stay
- Proportion charts (e.g., donut/pie) for booking channel share
- Table/matrix views for detailed breakdowns

23.4 Filters/Slicers

- Hotel Type
- Booking Channel
- Customer Type
- Time Period (Year/Month)
- Deposit/Payment Type

23.5 Drill-Down / Drill-Through Requirements

- Drill-down from yearly to monthly to daily trends
- Drill-through from summary KPI to detailed segment-level view

23.6 Business Questions Each Section Should Answer

Dashboard Section	Business Question Answered
Overview	What is our overall booking and cancellation performance?
Cancellation Analysis	Where are cancellations concentrated?
Customer Behaviour	How do customers behave differently across segments?
Trend & Seasonality	How does demand change over time?
Business Impact	What is the cost of cancellations to the business?

24. Excel Requirements

Excel may be used for:

- Initial data review and exploration
- Data validation and quality checks
- Basic data preparation (cleaning, structuring)
- Business-level summary checks before dashboard development
- Supporting ad-hoc business validation requests

Excel formulas themselves are not to be provided in this document.

25. SQL Requirements

SQL is expected to support:

- Data extraction from source systems/tables
- Data cleaning and structuring
- Aggregation of booking and cancellation data at required business dimensions
- Preparation of data models suitable for Power BI reporting

SQL queries themselves are not to be provided in this document.

26. Expected Deliverables

1. Cleaned and validated booking dataset (prepared, not created here)
 2. SQL-based data preparation logic (developed separately by Data Analyst)
 3. Excel-based validation/review documentation
 4. Power BI dashboard covering all required sections
 5. Business insights summary document (based on actual analysis, prepared later by Data Analyst)
 6. Documentation of assumptions, limitations, and data quality issues
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27. Business Acceptance Criteria

The project will be considered acceptable when:

- All business questions listed in Section 13 have been addressed
 - All KPIs listed in Section 15 are available and clearly defined in the dashboard
 - The Power BI dashboard is functional, accurate, and business-friendly
 - Data quality issues have been identified and documented
 - Stakeholders can independently interpret booking and cancellation patterns using the dashboard
 - No prediction, machine learning, or unapproved tools have been used
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28. Assumptions

- Historical booking data is available in a structured format suitable for analysis.
 - Data covers a sufficient time period to identify meaningful trends.
 - Only Excel, SQL, and Power BI will be used for this project.
 - The business requires descriptive/diagnostic insights, not predictive forecasts.
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29. Constraints

- Analysis must be limited to Microsoft Excel, SQL, and Power BI.
- No Machine Learning, AI, or advanced statistical modelling is permitted.
- No predictive analysis or forecasting is in scope.

- Analysis must remain at a business-understandable level of complexity.
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30. Risks and Dependencies

Risk/Dependency	Description
Data Availability	Analysis depends on availability of complete and accurate historical data
Data Quality	Poor data quality may limit reliability of findings
Tool Limitations	Descriptive-only tools may limit depth of certain analyses
Stakeholder Alignment	Business questions must remain aligned with stakeholder expectations
Scope Creep	Risk of expanding into predictive analysis beyond agreed scope

31. Success Criteria

- Clear identification of booking and cancellation patterns
 - A functional, business-friendly Power BI dashboard
 - Well-documented data quality and preparation process
 - Business stakeholders gain actionable understanding of cancellation behaviour
 - Deliverables completed using only approved tools (Excel, SQL, Power BI)
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32. Future Scope

- Potential future use of predictive analytics or Machine Learning (outside current scope)
 - Potential integration with real-time booking systems
 - Potential expansion to include customer satisfaction or review data
 - Potential development of automated alerting for high-cancellation-risk segments (future phase only)
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End of Document